

DATA ANALYSIS WITH PANDAS REPORT

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Domain :

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1. Objective

To demonstrate data analysis skills using Pandas by loading a CSV dataset, cleaning data, applying filtering, grouping, and aggregation techniques, and generating meaningful insights.

2. Requirements Covered

- Load and inspect a CSV dataset
- Clean missing or incorrect data
- Apply filtering techniques
- Perform grouping and aggregation
- Generate insights from the dataset
- Use Pandas for data analysis

3. Development Process (Step by Step)

Step 1 - Import Pandas Library

```
import pandas as pd
```

Step 2 – Load CSV Dataset

```
df = pd.read_csv("sales_data.csv")
```

```
response = requests.get(url)
```

Step 3 – Inspect Dataset

```
print(df.head())
```

```
print(df.info())
```

Step 4 – Clean Missing Values

```
df.dropna(inplace=True)
```

Step 5 – Filter Data

```
electronics = df[df["Category"] == "Electronics"]
```

Step 6 – Group and Aggregate

```
sales_summary = df.groupby("Category")["Sales"].sum()
```

Step 7 – Display insights

```
print(sales_summary)
```

4. Final Python Code

```
import pandas as pd

# Load CSV file

df = pd.read_csv("sales_data.csv")

print("=== Original Dataset ===")

print(df)

# Inspect dataset

print("\n=== Dataset Information ===")

print(df.info())

# Check missing values

print("\n=== Missing Values ===")

print(df.isnull().sum())

# Filter Electronics products

electronics = df[df["Category"] == "Electronics"]

print("\n=== Electronics Products ===")

print(electronics)

# Group by Category and calculate total sales

category_sales = df.groupby("Category")["Sales"].sum()
```

```

print("\n=== Total Sales by Category ===")

print(category_sales)

# Highest selling product

top_product = df.loc[df["Sales"].idxmax()]

print("\n=== Highest Selling Product ===")

print(top_product)

# Average sales

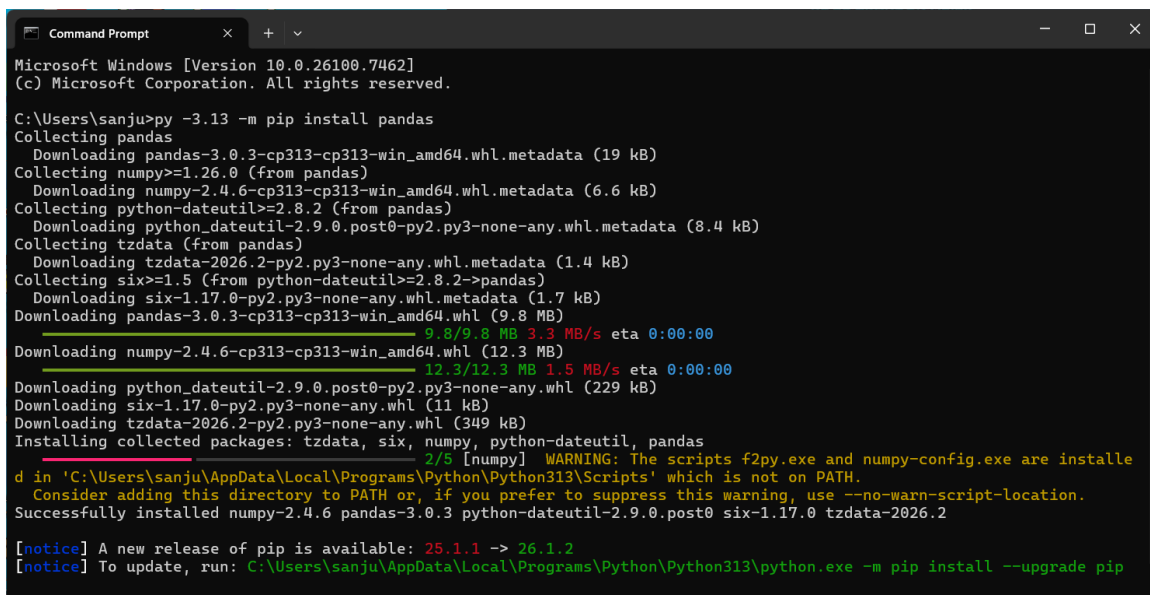
avg_sales = df["Sales"].mean()

print("\nAverage Sales:", avg_sales)
import requests

```

5. Screenshots with Captions

Figure 1 : Figure 1: Pandas Library Installation



```

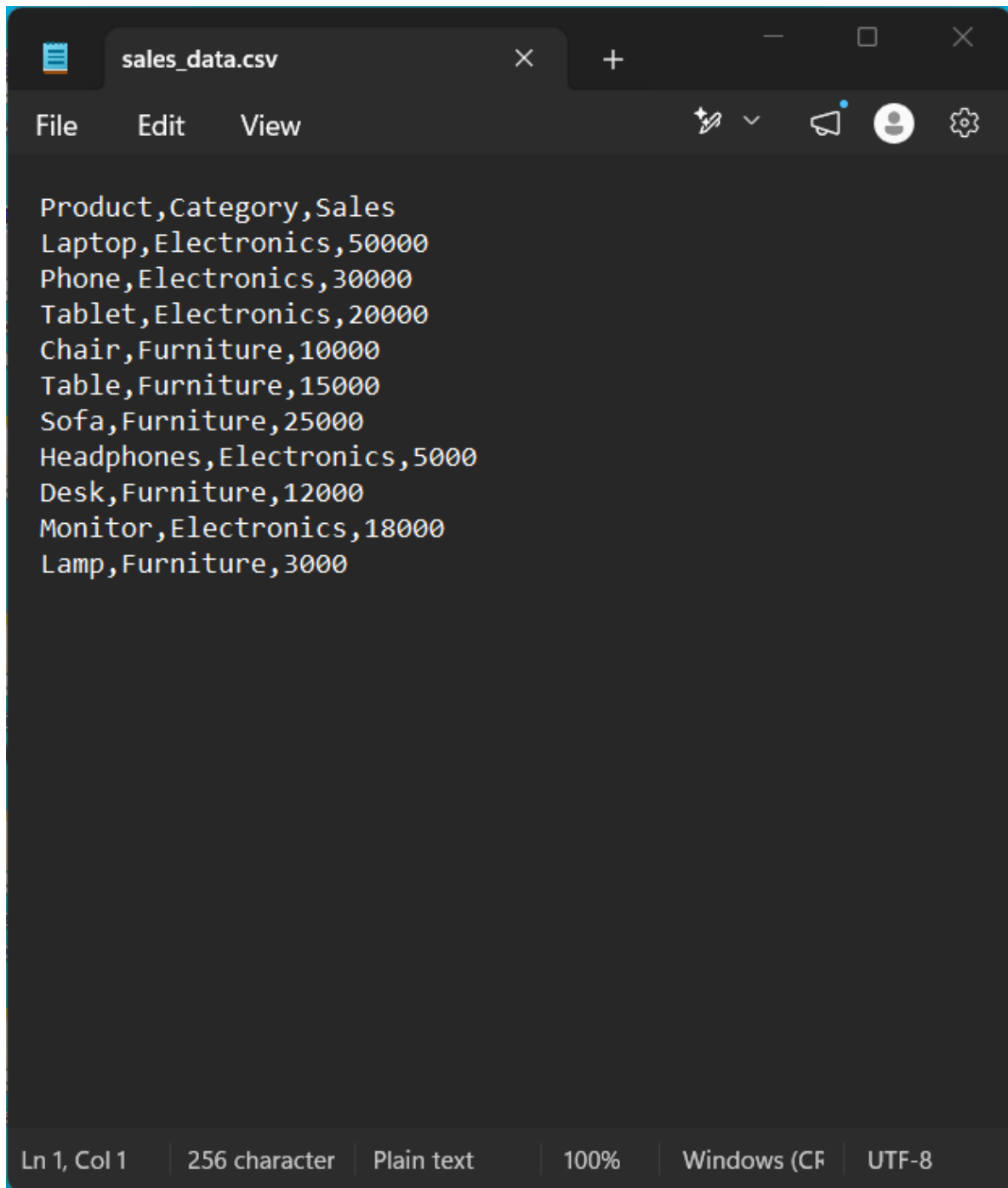
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sanju>py -3.13 -m pip install pandas
Collecting pandas
  Downloading pandas-3.0.3-cp313-cp313-win_amd64.whl.metadata (19 kB)
Collecting numpy>=1.26.0 (from pandas)
  Downloading numpy-2.4.6-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Downloading tzdata-2026.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-3.0.3-cp313-cp313-win_amd64.whl (9.8 MB)
  9.8/9.8 MB 3.3 MB/s eta 0:00:00
Downloading numpy-2.4.6-cp313-cp313-win_amd64.whl (12.3 MB)
  12.3/12.3 MB 1.5 MB/s eta 0:00:00
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, six, numpy, python-dateutil, pandas
  2/5 [numpy] WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\Users\sanju\AppData\Local\Programs\Python\Python313\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed numpy-2.4.6 pandas-3.0.3 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.2

[notice] A new release of pip is available: 25.1.1 -> 26.1.2
[notice] To update, run: C:\Users\sanju\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip

```

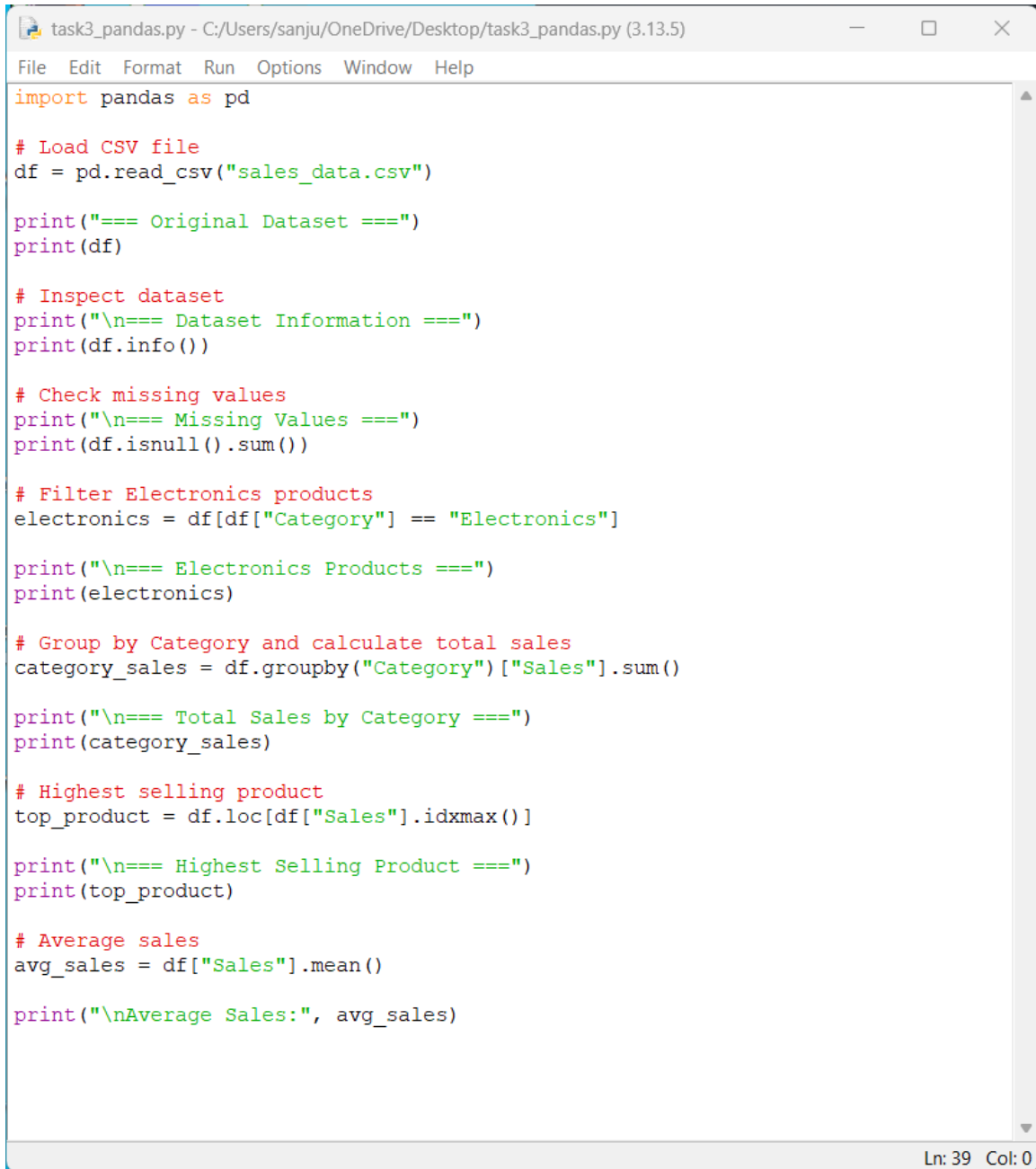
Figure 2 : Figure 2: Sales Dataset Creation (sales_data.csv)



The image shows a text editor window with a dark theme. The title bar at the top indicates the file is named 'sales_data.csv'. The menu bar includes 'File', 'Edit', and 'View'. The main text area contains a CSV dataset with 12 rows. The first row is the header: 'Product,Category,Sales'. The subsequent rows list various products and their sales figures. The status bar at the bottom shows 'Ln 1, Col 1', '256 character', 'Plain text', '100%', 'Windows (CF', and 'UTF-8'.

```
Product,Category,Sales
Laptop,Electronics,50000
Phone,Electronics,30000
Tablet,Electronics,20000
Chair,Furniture,10000
Table,Furniture,15000
Sofa,Furniture,25000
Headphones,Electronics,5000
Desk,Furniture,12000
Monitor,Electronics,18000
Lamp,Furniture,3000
```

Figure 3 : Figure 3: Pandas Data Analysis Program

The image shows a screenshot of a Python IDE window titled 'task3_pandas.py - C:/Users/sanju/OneDrive/Desktop/task3_pandas.py (3.13.5)'. The window has a menu bar with 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The main area contains Python code for data analysis using Pandas. The code imports Pandas as 'pd', loads a CSV file 'sales_data.csv' into a DataFrame 'df', and prints the original dataset. It then inspects the dataset by printing its information. Next, it checks for missing values using 'df.isnull().sum()'. The code filters for 'Electronics' products and prints the resulting DataFrame. It then groups the data by 'Category' to calculate total sales, prints the results, and identifies the highest-selling product by using 'df.loc[df["Sales"].idxmax()]'. Finally, it calculates the average sales across all categories and prints the result. The status bar at the bottom right indicates 'Ln: 39 Col: 0'.

```
task3_pandas.py - C:/Users/sanju/OneDrive/Desktop/task3_pandas.py (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd

# Load CSV file
df = pd.read_csv("sales_data.csv")

print("=== Original Dataset ===")
print(df)

# Inspect dataset
print("\n=== Dataset Information ===")
print(df.info())

# Check missing values
print("\n=== Missing Values ===")
print(df.isnull().sum())

# Filter Electronics products
electronics = df[df["Category"] == "Electronics"]

print("\n=== Electronics Products ===")
print(electronics)

# Group by Category and calculate total sales
category_sales = df.groupby("Category")["Sales"].sum()

print("\n=== Total Sales by Category ===")
print(category_sales)

# Highest selling product
top_product = df.loc[df["Sales"].idxmax()]

print("\n=== Highest Selling Product ===")
print(top_product)

# Average sales
avg_sales = df["Sales"].mean()

print("\nAverage Sales:", avg_sales)

Ln: 39 Col: 0
```

Figure 4 : Dataset Inspection Using Pandas

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:\Users\sanju\OneDrive\Desktop\task3_pandas.py =====
=== Original Dataset ===
   Product  Category  Sales
0   Laptop  Electronics  50000
1    Phone  Electronics  30000
2   Tablet  Electronics  20000
3    Chair   Furniture  10000
4    Table   Furniture  15000
5    Sofa    Furniture  25000
6 Headphones Electronics   5000
7     Desk    Furniture  12000
8   Monitor Electronics  18000
9     Lamp    Furniture   3000

=== Dataset Information ===
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Product     10 non-null     str
1   Category    10 non-null     str
2   Sales       10 non-null     int64
dtypes: int64(1), str(2)
memory usage: 372.0 bytes
None
```

Figure 5 : Final Data Analysis Output

```
=== Missing Values ===
Product      0
Category     0
Sales        0
dtype: int64

=== Electronics Products ===
   Product  Category  Sales
0   Laptop  Electronics  50000
1    Phone  Electronics  30000
2   Tablet  Electronics  20000
6 Headphones Electronics   5000
8   Monitor Electronics  18000

=== Total Sales by Category ===
Category
Electronics    123000
Furniture       65000
Name: Sales, dtype: int64

=== Highest Selling Product ===
Product      Laptop
Category     Electronics
Sales        50000
Name: 0, dtype: object

Average Sales: 18800.0

>>>
```

6. Insight Summary

The sales dataset was successfully analyzed using the Pandas library. The dataset contained 10 records with no missing values, indicating good data quality. After filtering the Electronics category, five electronic products were identified. Category-wise aggregation revealed that Electronics generated the highest total sales of **123,000**, while Furniture recorded total sales of **65,000**. The **Laptop** was identified as the highest-selling product with sales of **50,000**, and the average sales value across all products was **18,800**. These insights demonstrate how Pandas can be used to extract meaningful information and support data-driven decision-making.

7. Conclusion

This project successfully demonstrated data analysis using the Pandas library. The CSV dataset was loaded, inspected, and analyzed through filtering, grouping, and aggregation techniques. The analysis helped identify category-wise sales performance, the highest-selling product, and average sales values. The project fulfilled all the requirements and provided practical experience in data analysis and insight generation using Python.